Ukrainian Catholic University

AI Course · Semester Project Proposal · 2026

Benchmarking Medical Vision-Language Models for Radiology Image Captioning

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https://github.com/krasniukevych-marichka/Benchmarking-Medical-Vision-Language-Models-for-Radiology-Image-C

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1. Problem and motivation

The rapid increase in the volume of medical examinations, such as CT, MRI and radiography, creates an excessive workload for radiologists, leading to burnout and delays in preparing diagnostic reports. To address this problem, this project focuses on a comparative analysis of multimodal artificial intelligence models (Vision-Language Models) capable of automatically generating textual descriptions for radiological images. The key concept of this approach is utilizing AI as a digital assistant rather than replacing the physician or making autonomous diagnoses. The model generates a preliminary draft description, which the radiologist simply reviews and approves, significantly reducing the time spent on routine tasks.

2. Related Work and Existing Solutions

The study [1] compares ten general-purpose multimodal vision-language models of varying sizes on the ROCOv2 dataset, fine-tuned using LoRA, a method that adapts a model by updating less than 1% of its parameters. The results show that targeted fine-tuning of key layers is more effective than broader fine-tuning strategies and that larger models generally achieve higher accuracy than smaller ones, although some compact models come close in quality.

The paper [2] evaluates several ready-to-use large multimodal models (GPT 4o, Gemini 2.5 pro, Claude 4.5 Sonnet, LLaMA 4) on the same dataset without any additional fine-tuning, both in a zero-shot setting and with RAG, where the model is shown similar examples before generating a caption. RAG consistently improves caption quality across all metrics, with the best-performing configuration approaching the level of specialized fine-tuned models.

The overview paper [3] summarizes the results of the ImageCLEFmedical Caption 2026 challenge on medical concept detection and caption generation. The winning approaches combine ensembles of convolutional and transformer-based image classifiers (e.g., ConvNeXt-V2, BiomedCLIP, DenseNet) for concept classification with fine-tuned vision-language models (e.g., Gemma-3, Qwen-MedVL, LLaVA) for caption generation.

3. Data

This work utilizes the open-source multimodal radiological dataset **ROCOv2** (Radiology Objects in COntext v2) [4], constructed from medical scientific publications in PubMed Central. The dataset is distributed under the Creative Commons Attribution license (CC BY) for research and academic purposes.

The dataset contains a total of **79,789 radiological images**, split into three standard subsets:

- **Training set:** 59,958 images;
- **Validation set:** 9,904 images;
- **Test set:** 9,927 images.

Each image is annotated with medical concepts using a filtered target set (`*_concepts_manual.csv`) based on the Unified Medical Language System (UMLS CUI). This set contains 18 manually selected concepts representing 6 major radiological modalities (CT, X-ray, MRI, Ultrasound, Angiogram, PET), 8 anatomical regions in 4 different projections. The annotations follow a multi-label structure, as a single image can depict multiple concepts, resulting in 116,790 total annotations (an average of 1.46 concepts per image; 728 images do not contain target labels).

Additionally, natural language captions (`*_captions.csv`) are provided for all 79,789 images across all dataset splits (Train, Valid and Test), describing radiological findings and enabling multimodal analysis.

4. Approach and methods

We evaluate four pretrained medical vision-language models - MedGemma, LLaVA-Med v1.5, MedMO-4B-Next, and Hulu-Med-4B - on medical image captioning without fine-tuning. Each model receives the image and a fixed instruction to generate a one-sentence caption. To investigate whether additional medical con-

text improves the performance of pretrained VLMs, we augment the prompts with UMLS CUI concepts predicted by a separate multi-label detector. We consider two concept settings: a curated set of 18 CUIs covering mainly imaging modality and anatomical region, and the full vocabulary of 1,947 CUIs, which additionally includes medical findings. The concept detector is an ImageNet-pretrained ResNet-50 trained on the training split with asymmetric loss to address label imbalance, with small rotation and zoom augmentations. A global prediction threshold is tuned on the development split to maximize sample-averaged F1. We compare captioning performance with and without concept-based context to assess whether explicit medical information improves pretrained VLM generation.

5. Evaluation of Results

Quantitative evaluation of the generated descriptions is performed using four key metrics: **BLEU-1–BLEU-4** (measuring lexical precision from individual terms to sequential phrases), **ROUGE-L** (evaluating completeness and sentence structure), **CIDEr** (focusing on rare clinical findings via TF-IDF) and **BERTScore** (determining semantic similarity through contextual embeddings). Standard description generation by models using a basic prompt (without additional context) serves as the baseline.

6. Plan and Risks

Before midterm: data preparation and EDA of ROCOv2, training the baseline ResNet-50 concept detector on the full vocabulary (1,947 CUIs) and generating baseline captions from the four VLMs with the evaluation pipeline (BLEU, ROUGE-L, CIDEr, BERTScore). **Before the final:** narrowing detector predictions to the 18 manual CUIs, integrating predicted concepts into the VLM prompts, and running the full comparative analysis with conclusions. **Risks:** (1) limited compute for running four VLMs on the full 9,927-image test set – plan B: evaluate on a representative subset stratified by concept distribution; (2) class imbalance degrading detector quality on the full vocabulary – plan B: restrict predictions to the top-N most confident concepts or fall back to the 18-CUI manual set only.

7. Contribution statement

- **Viktoria Kukurudza** — 50%: Exploratory data analysis and preprocessing of the ROCOv2 dataset (UMLS CUI concepts, annotations, and text captions); development, data augmentation and training of the ResNet-50 concept detector using Asymmetric Loss and F1 threshold tuning; integration of predicted concepts into prompts.
- **Mariia Krasniukevych** — 50%: Configuration and inference of VLM models (MedGemma, LLaVA-Med v1.5, MedMO-4B-Next, Hulu-Med-4B); construction of an automated generation quality evaluation pipeline (BLEU-1–4, ROUGE-L, CIDEr, BERTScore); comparative analysis of results and documentation.

References

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